



Wireless magnetoelectric hydrogel spray reprograms the CXCL12-autophagy axis for spatiotemporally controlled cervical nerve root regeneration

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ABSTRACT

Cervical nerve root injury (CNRI) presents significant clinical challenges owing to complex anatomical constraints and poor intrinsic regenerative capacity. In this study, we introduce an *in situ*-sprayable magnetoelectric hydrogel (DG/FBD) engineered through dynamic covalent assembly and photo-triggered stabilization. The construction of critical boronate ester bonding between dopamine-modified Fe₃O₄@BaTiO₃ nanoparticles and phenylboronic acid-functionalized dextran endows DG/FBD with optimal viscosity (87 mPa s), rapid gelation (<2s) and robust tissue adhesion (22.11 kPa), allowing for precise deposition along tortuous nerve roots while conforming to branched cervical anatomy. We then establish a wireless magnetic actuation platform with conductivity modulation enabled spatiotemporally controllable bioelectronic interfacing. In CNRI models loading with magnetic stimulation, we found the hydrogel spray significantly promotes axonal regeneration and myelination, and importantly, improving functional recovery, including enhanced pain and temperature sensation, as well as increased grip strength. We also demonstrate accelerated neural differentiation of BMSCs and attenuated astrogliosis by *in vitro* studies. Mechanistically, we reveal that magnetoelectrically triggered CXCL12 upregulation drives Schwann cells remodeling through PI3K/AKT/mTOR inhibition and enhances autophagic flux, establishing a bioelectronic-metabolic bridge where the CXCL12-autophagy axis emerges as a fundamental regulatory node for neural regeneration. Collectively, our work pioneers wireless magnetoelectric regulation of chemokine-directed neural repair via an integrated sprayable hydrogel system that overcomes cervical interfacing constraints, establishing a promising platform for spatiotemporal microenvironmental reprogramming in complex neuropathies.

1. Introduction

Cervical nerve root injury (CNRI), resulting from compression or damage to the nerve roots in the cervical spine, manifests clinically as neck or shoulder pain, muscle weakness, sensory disturbances, and diminished deep tendon reflexes [1]. Intervertebral disc herniation is the most common cause of CNRI, with projections estimating nearly 990 million cases globally by 2050, posing a significant global health challenge [2,3]. Although surgical interventions alleviate compression,

many patients endure chronic symptoms requiring prolonged rehabilitation [4,5]. The deep, confined, and multi-branched anatomy of cervical nerve roots severely limits conventional neurorehabilitation, underscoring the urgent need for specific repair strategies.

Given that neural signals are transmitted as electrical impulses, electrical stimulation (ES) significantly enhances nerve regeneration [6–8]. However, existing transdermal or implanted ES systems often rely on rigid connections or battery support, complicating their operation and reducing patient compliance [9–11]. Magnetoelectric nanoparticles

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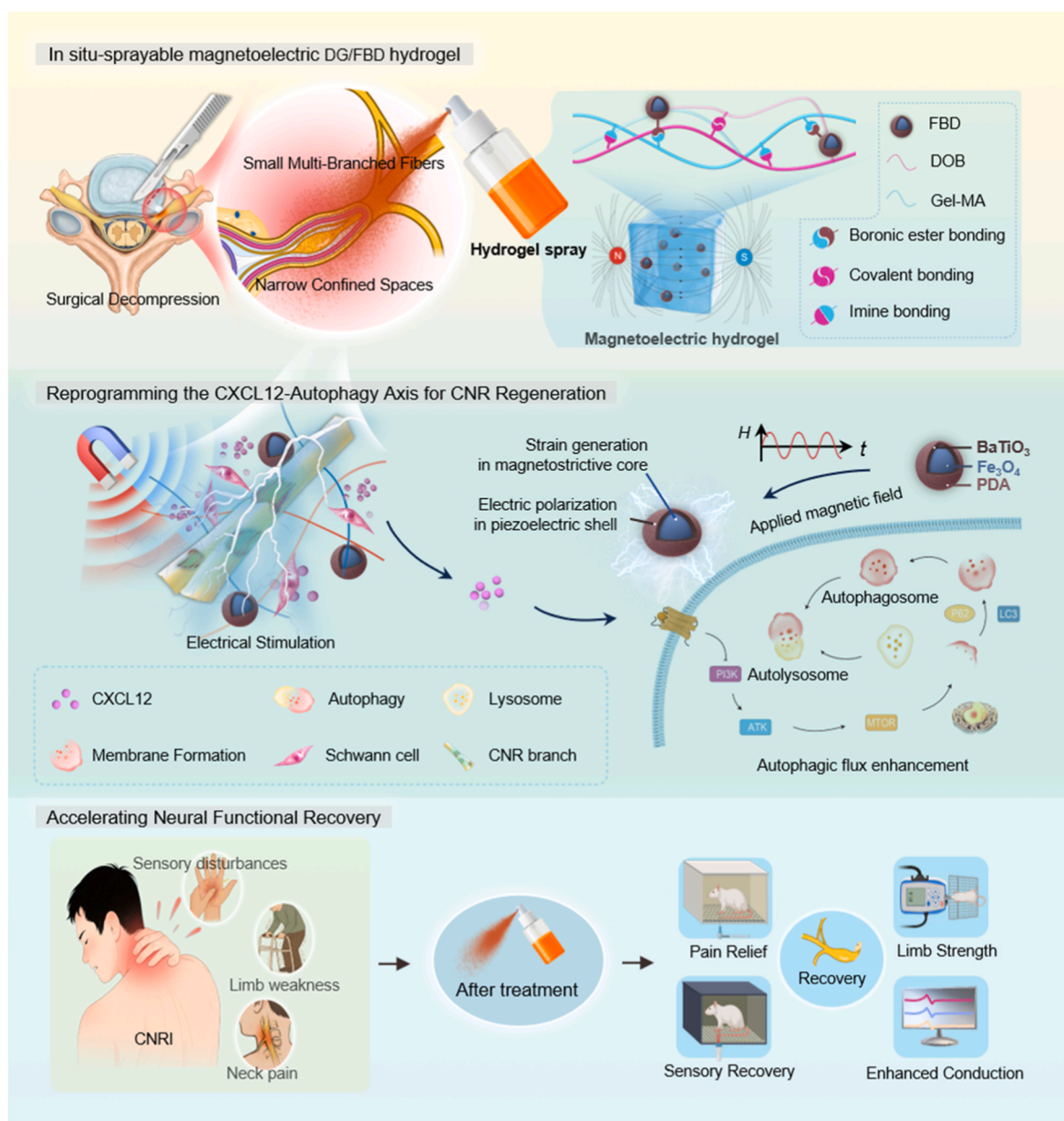
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(MENPs) offer a promising alternative. When remotely activated by magnetic fields, they generate localized electrical pulses near injured nerves, eliminating wires and improving precision. Since effective stimulation requires micrometer-scale proximity to target cells, secure anchoring of MENPs is critical. As a result, bioactive hydrogels that mimic the native microenvironment are attracting intense interest as idea carriers for in situ MENP delivery and fixation [12,13]. In most cases, these composite hydrogels are shaped into conduits or patches that wrap around the injured nerve, generating an electric field to alleviate nerve inhibition and promote neuroprotection [14,15]. However, their application to deep and confined cervical roots remains impractical, necessitating materials with in situ branch permeability and adaptive adhesion.

To overcome these limitations, as shown in Scheme 1, we propose an in situ-sprayable magnetoelectric hydrogel system for CNRI repair after foraminal decompression. The system comprises: (A) phenylboronic acid-grafted oxidized dextran (PBA-OD) complexed with dopamine-

modified $\text{Fe}_3\text{O}_4@/\text{BaTiO}_3$ (FBD), and (B) methacrylamide gelatin (MG). Alternating A/B spray cycles enable rapid sol-gel transition via imine crosslinking and hydrogen bonding, and the subsequent photopolymerization stabilizes the network into a uniform, adhesive hydrogel layer (DG/FBD, a composite of functionalized dextran, gelatin and FBD). This sprayable, morphologically adaptive design ensures precise deposition and conformal coverage of branched cervical roots. Crucially, FBD with dopamine modification enhances nanoparticle dispersion and facilitates their covalent bonding with PBA-OD via boronic ester formation. Meanwhile, rapid in situ gelation ensures conformal adhesion and tight fixation of the magnetoelectric hydrogel spray to the nerve surface. Under remote magnetic actuation, the DG/FBD hydrogel generates localized electrostimulation, promoting nerve regeneration and facilitating the recovery of sensory and motor functions in rats.

While ES is widely recognized to accelerate peripheral nerve regeneration via calcium-dependent MAPK/ERK/PI3K cascades and neurotrophic secretion, pivotal questions remain unanswered [16,17].



Scheme 1. Schematic of DG/FBD hydrogel spray design synergized with remote electrical stimulation to orchestrate CNRI repair and neural functional recovery. Mechanistically, magnetoelectrically triggered CXCL12 upregulation initiates Schwann cell remodeling through PI3K/AKT/mTOR pathway inhibition and autophagic flux potentiation, bridging bioelectronic signaling with metabolic clearance.

Most notably, the specific mechanisms of magnetoelectric stimulation in neural repair are largely unexplored. Equally overlooked is whether ES rescues impaired post-injury autophagic flux, a lysosomal degradation process essential for cellular homeostasis [18]. Following neural injury, compromised autophagic clearance exacerbates cellular damage and impedes regenerative processes, whereas restoration of autophagic flux has shown potential to enhance neuronal survival [19]. Building on this platform, we pioneer the discovery that magnetoelectrically triggered CXCL12 upregulation initiates Schwann cell remodeling through PI3K/AKT/mTOR pathway inhibition and autophagic flux potentiation, bridging bioelectronic signaling with metabolic clearance. Our biocompatible, adaptive strategy thus overcomes cervical interfacing constraints while mechanistically linking ES to autophagy regulation, establishing the first evidence of magnetoelectric material-mediated chemokine regulation in CNRI repair.

2. Results and discussion

2.1. Synthesis and magnetoelectric properties of core/shell structured $\text{Fe}_3\text{O}_4@\text{BaTiO}_3@\text{PDA}$ nanoparticles (FBD)

Core/shell $\text{Fe}_3\text{O}_4@\text{BaTiO}_3@\text{PDA}$ nanoparticles (FBD, ~ 300 nm, Fig. S1) were synthesized via a sequential process, involving sol-gel deposition of TiO_2 (TO) on Fe_3O_4 (FO) to form $\text{Fe}_3\text{O}_4@\text{TiO}_2$ (FT), hydrothermal conversion of Ba^{2+} to form $\text{Fe}_3\text{O}_4@\text{BaTiO}_3$ (FB), and subsequent alkaline coating with PDA (Fig. 1A). Transmission electron microscopy (TEM) (Fig. 1B) and field emission scanning electron microscope (SEM) (Fig. S2) confirmed uniform encapsulation of 25-nm PDA layer, with energy dispersive X-ray (EDX) mapping showing homogeneous elemental distribution (Fig. 1C). Successful PDA functionalization was further evidenced by an increased negative zeta potential (Fig. 1D) and characteristic shifts in FT-IR peaks (Fig. 1E), establishing critical bonding capacity for hydrogel integration. Following structural confirmation by X-ray Diffraction (XRD), which verified that the Fe_3O_4

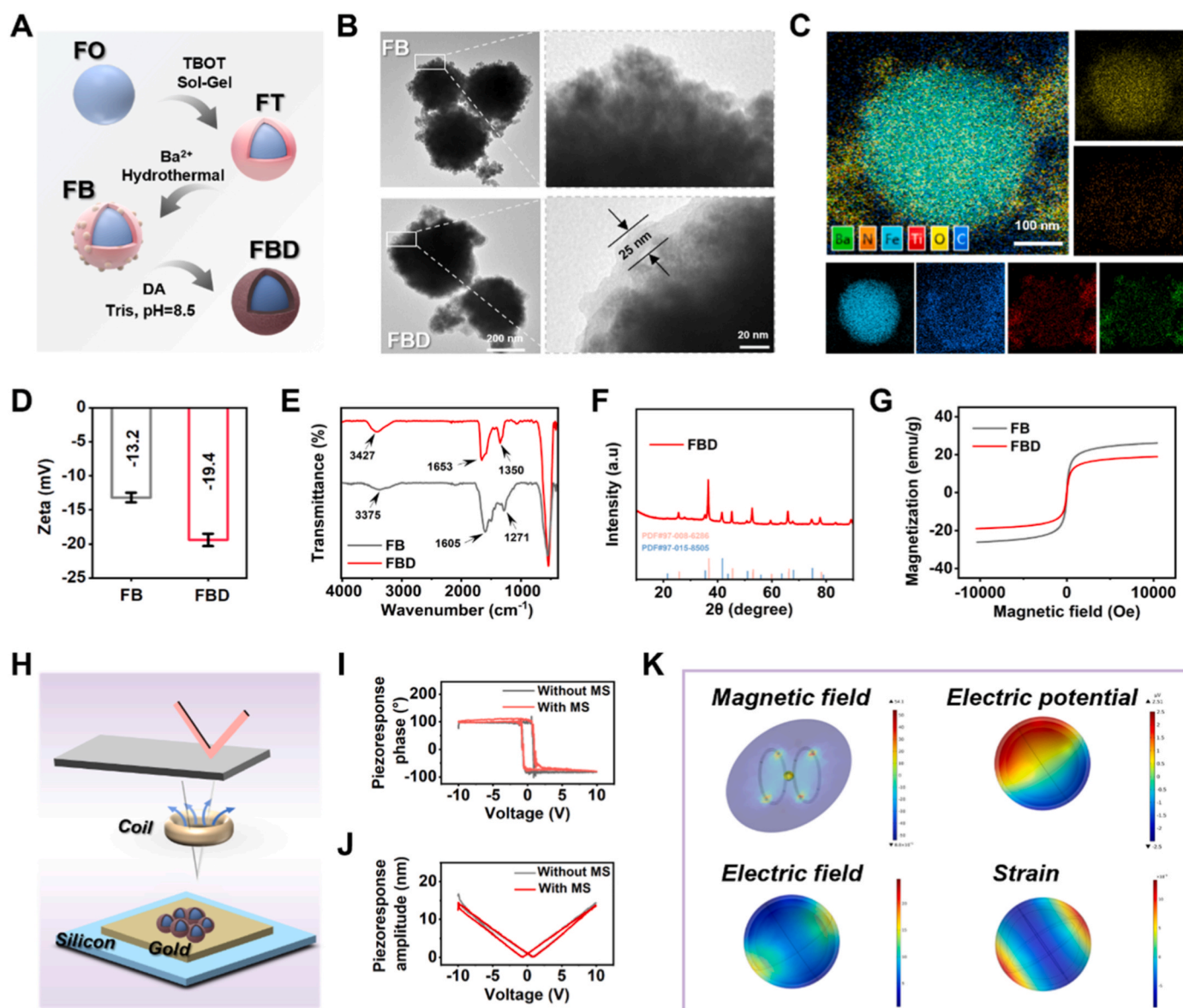


Fig. 1. Synthesis and characterization of dopamine-modified $\text{Fe}_3\text{O}_4@\text{BaTiO}_3$ (FBD) nanoparticles. (A) Schematic synthesis of FBD nanoparticles. (B) TEM images. (C) EDX elemental mapping. (D) Zeta potential. (E) FT-IR spectrum. (F) XRD. (G) Magnetic hysteresis loop. (H) Schematic illustration of measuring magnetoelectric effects via AFM. (I) Amplitude-voltage and (J) Piezoelectric phase response curves under 1000 Oe magnetic field. (K) Finite element analysis of the magnetoelectric effects under 15 mT (Electric potential [μV], electric field [V/m], strain [ppm]).

cubic phase (PDF#97-015-8505) and BaTiO₃ tetragonal phase (PDF#97-008-6286) were preserved in FBD (Fig. 1F), we then evaluated their functional properties. Importantly, vibrating sample magnetometer (VSM) confirmed that the superparamagnetic properties of the nanoparticles were retained after PDA coating, with a magnetic saturation value of 18 emu·g⁻¹ (Fig. 1G). Simultaneously, piezoresponse force microscopy (PFM) analysis verified the functional integrity of the nanoparticles (Fig. 1H). The 180° phase shifts (Fig. 1I) demonstrated the polarization switching capability, while the butterfly-shaped strain curves (Fig. 1J) confirmed the reversible polarization behavior essential for magnetoelectric conversion. Under magnetic stimulation (1000 Oe),

the coercive voltages of BTO shifted from -0.59/0.86 V to -0.51/1.02 V, confirming effective magnetostrictive-piezoelectric coupling. Finite element analysis (Fig. 1K) further established that PDA coating caused no magnetoelectric degradation, with FBD generating equivalent 2.5 μV potentials to FB under 15 mT fields. These results collectively confirm that PDA modification preserves the magnetoelectric conversion efficiency essential for wireless neuromodulation while providing interfacial binding sites for subsequent hydrogel integration.

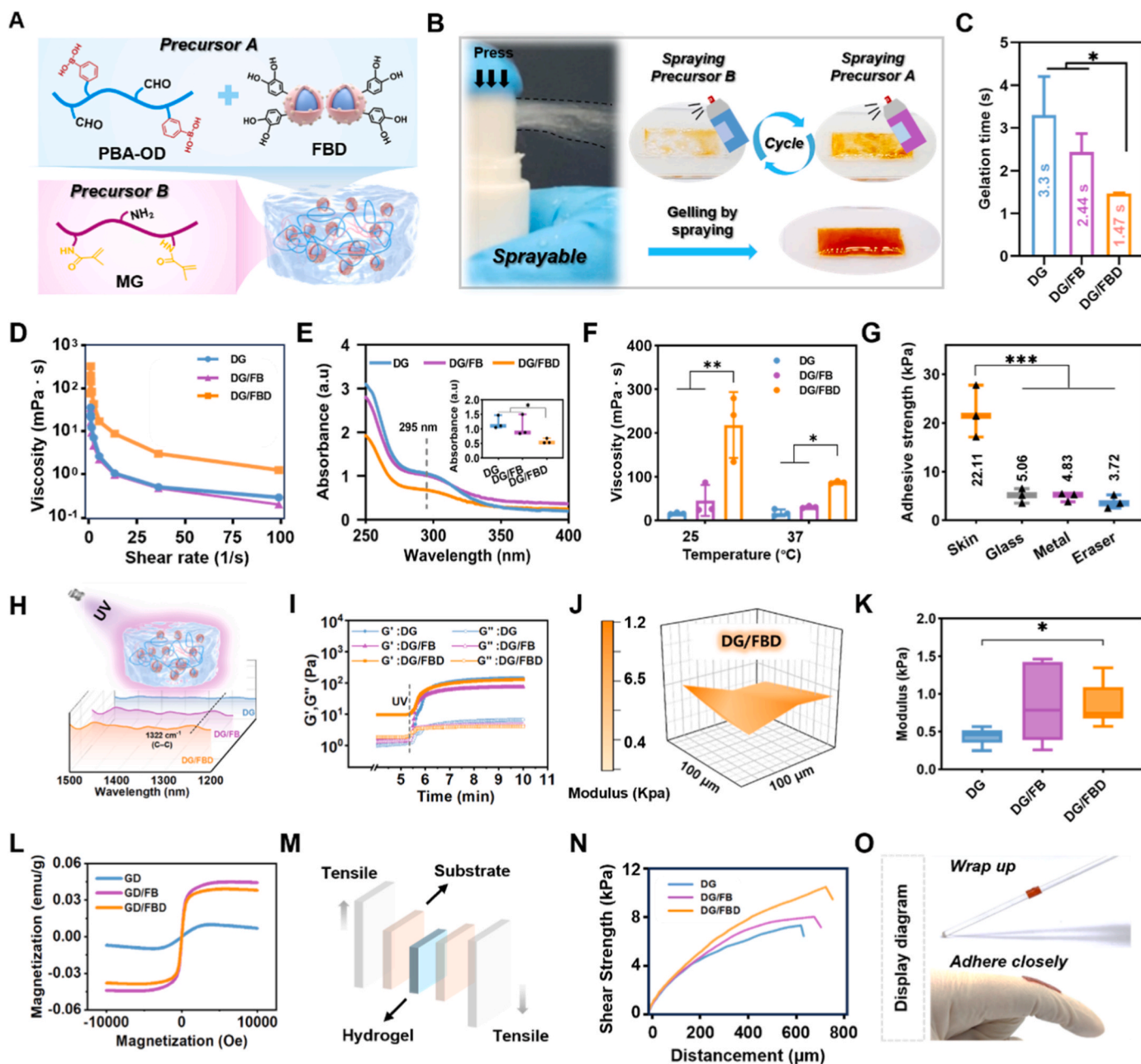


Fig. 2. Fabrication and characterization of *in situ*-sprayable magnetoelectric hydrogel (DG/FBD). (A–G) Pre-crosslinking: (A) Two-step spraying protocol: Spray A (PBA-OD complexed with FBD); Spray B (methylacrylated gelatin, MG). (B) Cyclic sequential deposition process. (C) Gelation time. (D) Viscosity at various shear rates. (E) The UV-Vis absorption at 295 nm. (F) Viscosity at different temperatures. (G) Adhesion strength on diverse substrates (pork skin, metal, glass, plastic). (H–O) Post-photocrosslinking: (H) FT-IR spectra (illustration above: UV-induced covalent crosslinking). (I) Rheological analysis of the crosslinking process. (J) Micromechanical modulus mapping by nanoindentation. (K) Quantitative modulus analysis. (L) Magnetic hysteresis loop. (M) Shear adhesion test schematic. (N) Shear adhesion strength quantification. (O) Adaptive adhesion behavior under curled tube and finger deformation. The data are presented as the mean ± SEM (**p* < 0.05, ***p* < 0.001).

2.2. Sprayable magnetoelectric hydrogel design for neural adaptation

To address the challenges of CNRI repair, an in situ-sprayed adhesive magnetoelectric hydrogel (DG/FBD) is tailored by integrating phenylboronic acid-functionalized oligodopamine (PBA-OD), methacrylated gelatin (MG), and FBD nanoparticles (Fig. 2A). The successful synthesis of MG and PBA-OD was confirmed by ^1H NMR and FTIR spectroscopy (Figs. S3–S4). To ensure clinical applicability, we developed an optimized two-step spraying protocol that prevented nozzle clogging while enabling uniform, layered adhesive deposition through sequential spraying cycles (Fig. 2B and S5). Its rapid gelation within less than 2 s enables immediate tissue fixation (Fig. 2C and S6). Additional rheological tests under physiologically relevant conditions confirmed reproducible rapid gelation and stable shear-thinning behavior of DG/FBD hydrogel, indicating its suitability for surgical application (Fig. S7). Rheological analysis revealed that the DG/FBD hydrogel's non-Newtonian behavior, which characterized by shear thinning and a consistently maintained moderate viscosity of 87 mPa s, providing smooth sprayability (Fig. 2D). Meanwhile, in contrast, the control hydrogels DG and DG/FB, lacking sufficient boronate ester crosslinking, display lower viscosities (17 mPa s and 37 mPa s) and slower gelation (3.3 s and 2.44 s), compromising post-spray fixation stability (Fig. 2D and S8). The critical role of boronate ester bonding was further confirmed through elevated viscosity in shear rate-dependent curves under mildly alkaline conditions (Fig. S9), with further validation from reduced UV–Vis absorption at 295 nm (0.58) compared to that of DG (1.21) and DG/FB (1.08) as illustrated in Fig. 2E. Additionally, the DG/FBD hydrogel maintained exceptional operational stability with temperature-resistant viscosity (Fig. 2F) and tissue-conformal adhesion during dynamic deformation (Fig. S10). Notably, the DG/FBD hydrogel exhibits robust interfacial adhesion to skin tissue (22.11 kPa), which may be attributed to the synergistic effects of hydrogen bonds, electrostatic interactions, and cation- π forces (Fig. 2G–S11 and S12).

Post-spray UV crosslinking reinforced the matrix via carbon–carbon (C–C) bond formation (1322 cm^{-1} FT-IR peak, Fig. S13 and 2H), elevating the storage modulus to neural-matching levels (~ 100 Pa, Fig. 2I) while preserving self-healing capability (Fig. S14). Nano-indentation quantified micromechanical homogeneity, with DG/FBD exhibiting an elastic modulus (0.21 kPa) intermediate between DG (0.08 kPa) and DG/FB (0.41 kPa) (Fig. 2J and K and S15). Functionally, the vibrating sample magnetometer (VSM) confirmed well-retained superparamagnetism in DG/FBD (Fig. 2L), maintaining the saturation magnetization essential for non-invasive neuromodulation. In addition, this spatiotemporally controlled system overcomes cervical anatomy constraints through instant adhesion and branch penetration capabilities (Fig. 2M – O and S16), establishing an adaptive platform for complex deep nerve injury repair. To further validate neural-tissue adhesion, the hydrogel was sprayed onto the exposed cervical nerve root, rinsed with PBS, and examined via H&E staining, demonstrating a continuous, tightly adherent layer without detachment (Fig. S17). Further quantitative assessment in Fig. S18 revealed uniform and conformal surface coverage, with hydrogel thickness ranging from 7.52 to 10.32 μm and averaging $8.82 \pm 1.06\text{ }\mu\text{m}$. Given the small diameter of the cervical nerve root ($\sim 245\text{ }\mu\text{m}$), this minimal thickness variation demonstrates highly coating uniformity and conformal coverage. Collectively, these data establish that the DG/FBD hydrogel adheres rapidly in complex anatomical locations and maintains stable integration with neural tissue under wet, dynamic conditions, offering a robust and practical platform for CNRI repair.

2.3. DG/FBD modulates neural differentiation and maturation of BMSCs *In vitro*

To ensure optimal biocompatibility and magnetoelectric stimulation efficacy, we screened an appropriate doping concentration of FBD nanoparticles (NP) for the hydrogel spray by co-culturing with BMSCs

and PC12 cells, respectively (Fig. S19A). Cell viability test revealed that NP concentrations below 0.5 % showed little inhibition on the proliferation of BMSCs, whereas a higher concentration of 1 % with visible suppression (Fig. S19B). Additionally, cell spreading of PC12 cells illustrated that magnetic field loading effectively stimulated axonal extension, with cellular elongation positively correlated with NP concentration (Fig. S19C–F). Based on these findings, 0.5 % NP was selected for subsequent studies to balance biocompatibility and magnetoelectric stimulation efficiency. We next evaluated the impact of DG/FBD hydrogel on the neuronal differentiation and maturation of BMSCs under magnetoelectric stimulation (denoted as + MS). On day 7, Tuj1 (an early neuronal marker [20]) expression was significantly upregulated in the DG/FBD + MS group compared to that of DG, DG + MS, and DG/FBD groups, indicating enhanced early neural differentiation (Fig. 3A and E). Similar upregulation trends were observed for neurofilament (NF, Fig. 3B and F) and postsynaptic density protein-95 (PSD95, Fig. 3C and G) after 21 days, which respectively indicate neuronal maturation and synaptic reconstruction [21,22]. Conversely, GFAP expression, a marker of astrocyte differentiation, was significantly suppressed in the DG/FBD + MS group, reflecting the inhibitory effect of magnetoelectric regulation on astrocyte activation and glial scar formation (Fig. 3D and H). These results collectively demonstrate that DG/FBD hydrogel-mediated magnetoelectric stimulation promotes BMSCs neuronal differentiation while suppressing glial lineage commitment.

2.4. DG/FBD hydrogel spray accelerates nerve regeneration in CNRI rats

To further validate its positive effect on the structural and pathophysiological repair of nerve roots, DG/FBD was sprayed in situ on rat CNRI lesions (Fig. 2I), and a magnetic field (15 mT, 60 Hz, duty cycle of 50 %) was applied to the sprayed rats (Fig. S20). Experimental groups (CNRI, DG + MS, DG/FBD, DG/FBD + MS, Sham) were established using standardized surgical procedures. Due to the narrow anatomical location of the cervical nerve roots, it was not feasible to clearly display how the DG/FBD hydrogel spray uniformly coats and adheres to the small nerve branches (Fig. S17B). Therefore, we selected the branched nerve endings of the sciatic nerve for demonstration. As shown in Fig. S21, the DG/FBD hydrogel spray uniformly coated and adhered to the intricate branches at the distal end of the sciatic nerve. After 4 weeks of treatment, H&E staining revealed sparse and hollow axons in CNRI, DG + MS, and DG/FBD groups, whereas DG/FBD + MS displayed dense and well-organized axonal fibers that closely resembling the structure of the Sham group (Fig. 3J and S22A). TEM analysis of myelin regeneration and quantitative assessment of nerve fiber parameters (myelin thickness, axon diameter, and G-ratio) further supported these findings. DG/FBD + MS showed thick and compact myelin surrounding well-organized axons, contrasting with thin and sparse myelin in control groups (Fig. 3K and S22B). Statistical analysis demonstrated significantly increased myelin thickness (Fig. 3L) and near-normal axon diameter in DG/FBD + MS (Fig. 3M), reinforcing the hydrogel's role in promoting nerve regeneration. G-ratio analysis, defined as the ratio of axonal diameter to the total myelinated fiber diameter [23], further confirmed optimal axonal functionality and structural integrity in DG/FBD + MS (Fig. S23). Longitudinal nerve section immunofluorescence staining (Fig. 3N & S24A–C) demonstrated significant upregulation of structural proteins in DG/FBD + MS, such as NF200 (axon marker [24]), MBP (myelin sheath marker [25]), and S100 (Schwann cell marker [26]). These results indicate that magnetoelectrically stimulated DG/FBD hydrogel spray accelerates structural nerve regeneration in CNRI rats.

2.5. DG/FBD hydrogel spray enhances functional recovery of sensory, motor, and nerve impulse conduction in CNRI rats

CNRI often lead to sensory abnormalities, such as pain and altered

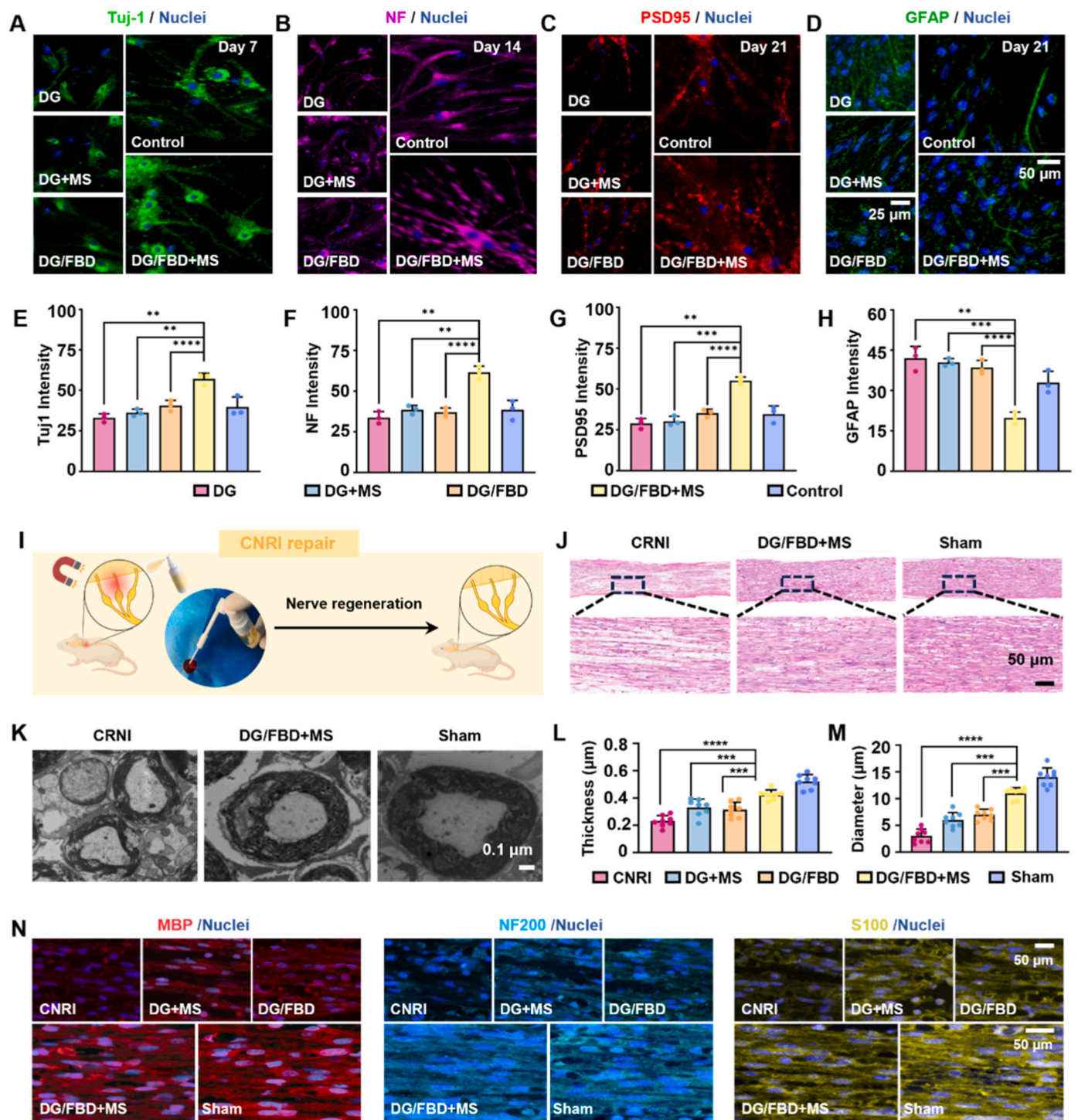


Fig. 3. Magneto-electric hydrogel regulates BMSC neural differentiation in vitro and enhances CNRI repair in vivo. (A–I) In vitro regulation of BMSC differentiation. (A–D) Immunofluorescence staining of neural markers: Tuj1 (7 days), NF (14 days), PSD95 (21 days), GFAP (21 days). (E–H) Quantified analysis of marker expression intensities. (I–N) In vivo CNRI repair assessment 4 weeks post-surgery. (I) Schematic diagram of rat CNRI model. (J) H&E staining of CNR tissue. (K) TEM images of the myelinated axons from the middle segment of regenerated CNR. (L–M) Statistical analysis of myelin thickness and axon diameter. (N) Immunofluorescence staining of MBP (red), NF200 (cyan), S100 (yellow). The data are presented as mean $\pm$ SEM (** p < 0.01, *** p < 0.001 and **** p < 0.0001). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

temperature perception, in the affected skin areas [27]. Hence, recovery of sensory function is a critical indicator of post-injury functional restoration [20,21]. To evaluate the potential of magnetic field-driven DG/FBD hydrogel spray in enhancing sensory recovery, we conducted Hargreaves and von Frey tests. In the Hargreaves test, rats treated with DG/FBD + MS exhibited significantly lower thermal pain thresholds at

week 2, representing a faster recovery of thermal sensitivity. By week 4, the thermal pain threshold in the DG/FBD + MS group approached that of the Sham group, indicating accelerated recovery of thermal sensation (Fig. 4A). In the von Frey test, the DG/FBD + MS group showed the most significant improvement, with pain threshold recovery approaching Sham levels by week 6, suggesting expedited recovery of pain sensitivity

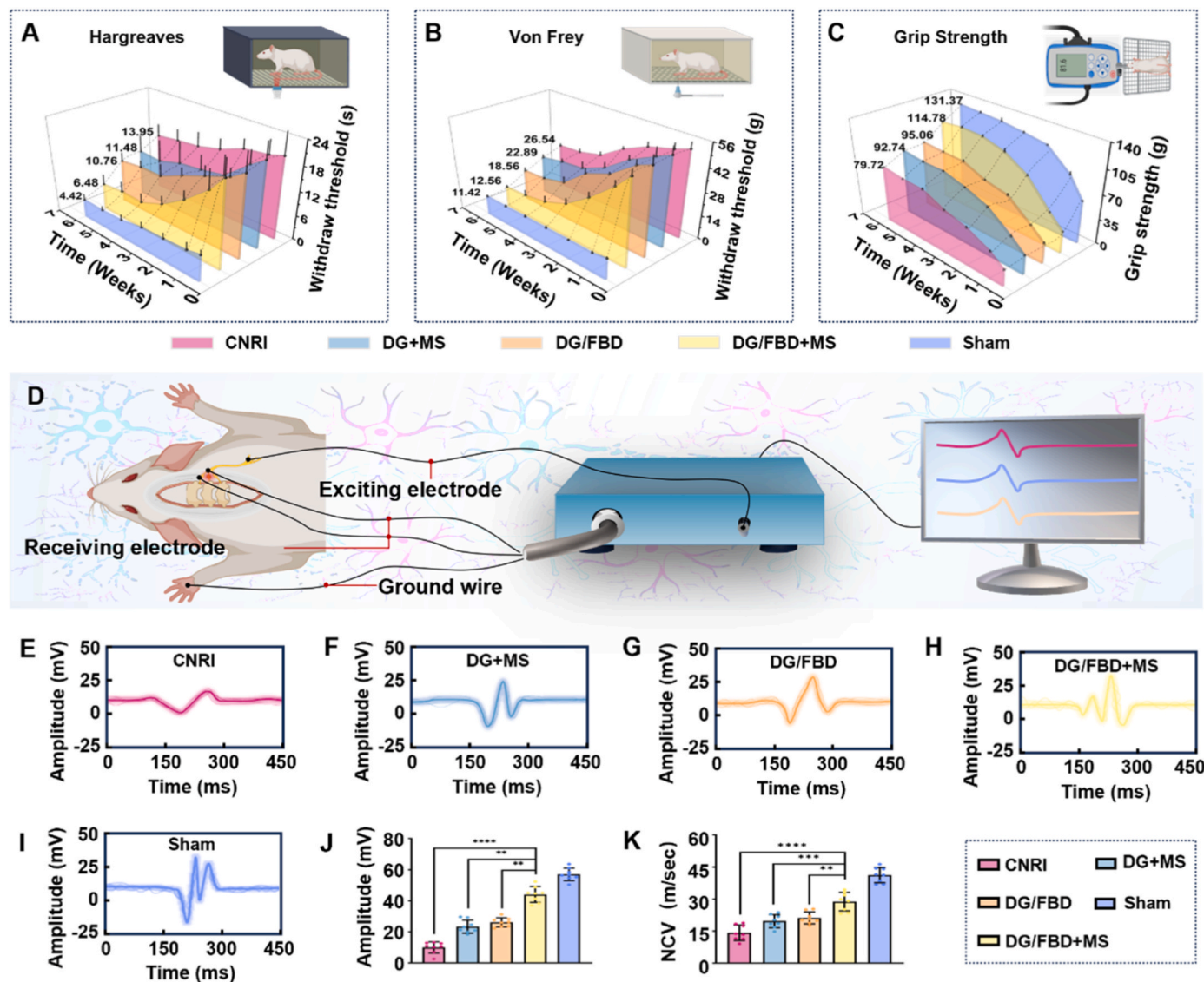


Fig. 4. Functional recovery assessment of CNRI rats post-surgery. (A) Hargreaves thermal pain test. (B) Von Frey mechanical sensitivity test. (C) Grip strength motor evaluation. (D) Experimental design for electrophysiological analysis. (E–I) Representative CMAP traces. (J) CMAP amplitude quantification. (K) Nerve conduction velocities (NCVs). The sample size of each group was greater than 8 rats. The independent experiment was repeated three times and then the average value was taken. The data are presented as mean $\pm$ SEM (** p < 0.01, *** p < 0.001 and **** p < 0.0001).

(Fig. 4B). To further assess motor function recovery, we performed Grip Strength Tests to evaluate limb mobility and muscle strength. Throughout the 6-week study period, grip strength in DG/FBD + MS-treated rats progressively recovered, reaching levels closest to the Sham group by week 4 (Fig. 4C). As shown in the schematic of Fig. 4D, electrophysiological analysis was also conducted to evaluate nerve conduction recovery. Specifically, compound muscle action potential (CMAPs) were recorded by exposing the cervical nerve root and stimulating it with a single pulse (Fig. 4E–I). As shown in Fig. 4J, the CMAP amplitude on the injured side in the DG/FBD + MS group was significantly higher than that in the unstimulated DG/FBD group and the DG + MS group, and close to that in the sham group. In contrast, CMAP amplitude remained significantly lower in the CNRI group than in the other treatment groups due to limited functional recovery. Furthermore, nerve conduction velocity (NCV) increased from 13.69 ± 6.84 m/s in the CNRI group to 41.78 ± 8.29 m/s in the DG/FBD + MS group, showing most substantial improvement among treatments (Fig. 4K). In summary, the enhanced CMAPs amplitude and NCV values in the DG/FBD + MS group reflect improved nerve fiber regeneration and

motor response [22,28]. These results confirm that the magneto-electrically stimulated DG/FBD hydrogel spray promotes comprehensive functional recovery in CNRI, including sensory and motor functions, as well as nerve impulse conduction. The innovation of this approach lies in the multiple advantages of the magneto-electric hydrogel spray. It not only offers excellent biocompatibility and magneto-electric responsiveness but also demonstrates outstanding anatomical adaptability, allowing for better conformation to tissues and efficient stimulation.

2.6. Magneto-electric stimulation enhances autophagic activity and upregulates CXCL12 expression

RNA transcriptome sequencing on tissues sampled from CNRI regeneration sites 4 weeks post-surgery was then performed to elucidate the mechanisms underlying nerve regeneration promoted by the magnetic field-driven DG/FBD hydrogel. The results revealed distinct clustering of samples by group with significantly different gene expression profiles (Fig. 5A), confirming the usability of sequencing data. Comparative analysis between the Sham and DG/FBD + MS groups

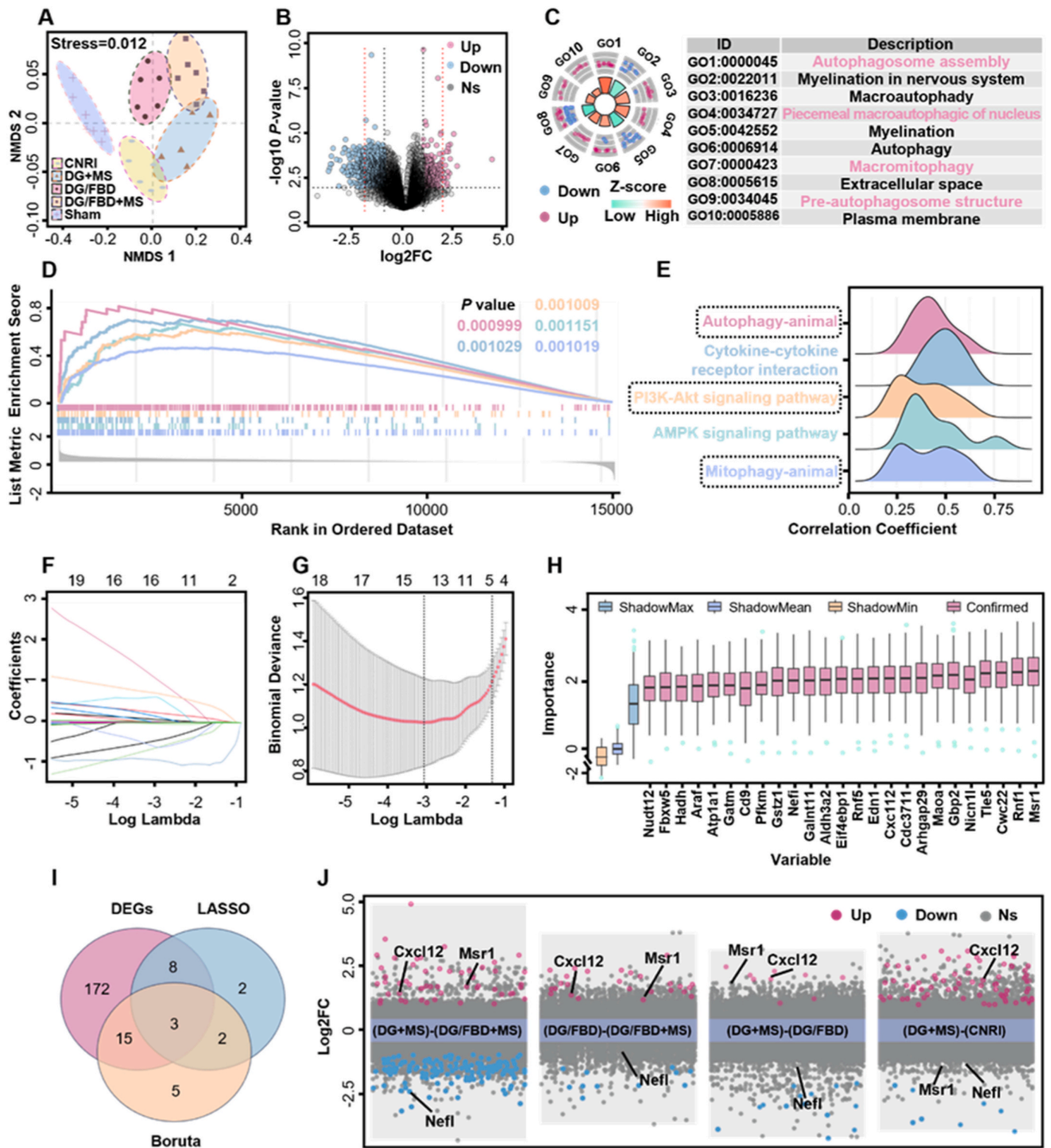


Fig. 5. RNA sequencing analysis after 4 weeks of CNRI repair. (A) NMDS plot. (B) DEG volcano plot (Red: upregulated; Blue: downregulated). (C) GO enrichment analysis and differential gene distribution (Red and blue dots respectively denote up- and down-regulated genes within each enriched GO term). (D–E) GSEA pathways. (F) Cross-validation of results of LASSO regression analysis. (G) LASSO regression biomarker screening. (H) Boruta algorithm-selected features. (I) Venn diagram of overlapping genes. (J) Distribution of differentially expressed genes and biomarker among different experimental groups. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

identified 126 upregulated and 212 downregulated genes (Fig. 5B). Gene Ontology (GO) enrichment analysis indicated that every term contained differentially expressed genes in varying proportions (with red and blue dots representing up- and down-regulated genes within each GO term, respectively). Notably, several autophagy-related

pathways were prominently enriched, including Autophagosome assembly, Piecemal macroautophagic of nucleus, Macroautophagy, and Pre-autophagosomal structure membrane (Fig. 5C). Additionally, Gene Set Enrichment Analysis (GSEA) indicated significant enrichment in autophagy-related pathways such as Autophagy-animal, Mitophagy-

animal, and PI3K-Akt signaling (Fig. 5D and E). These findings suggest that autophagy may play a critical role in the promotion of CNRI repair by magnetoelectric stimulation mediated by DG/FBD hydrogel. Subsequently, LASSO and Boruta algorithms were applied to identify key regulatory genes by selecting 15 and 25 feature variables, respectively (Fig. 5F–H). Nefl, Msr1, and Cxcl12 are identified as overlapping genes (Fig. 5I). Relative expression level analysis showed consistent upregulation of Cxcl12 in the DG/FBD + MS group, aligning with observed improvements in CNRI repair. In contrast, Nefl and Msr1 did not exhibit similar trends (Fig. 5J). These multi-omics analyses indicate that magnetoelectric stimulation mediated by DG/FBD hydrogel enhanced autophagy activity and up-regulated Cxcl12 expression. Previous studies have shown that, following peripheral nerve injury, nerve cells use autophagy to clear myelin debris in the Wallerian degeneration area, thereby creating space for nerve regeneration [29,30]. Additionally, the

CXCL12-CXCR4 axis has been implicated in the repair of damaged sciatic nerves [31]. However, this study offers a novel insight by demonstrating that the magnetoelectric stimulation not only enhances autophagy activity but also upregulates Cxcl12 expression, a finding not previously reported in the context of nerve electric modulation.

To validate these findings, we subsequently conducted relevant experiments (Fig. 6A). Quantitative PCR (qPCR, Fig. 6B–D) and enzyme-linked immunosorbent assay (ELISA, Fig. 6E) collectively corroborated that CXCL12 expression was significantly higher in the DG/FBD + MS group compared to other experimental groups. CXCL12, also recognized as Stromal Cell-Derived Factor-1A (SDF-1A), is a chemokine pivotal for neuronal migration and tissue repair [32]. It is typically upregulated following nerve injury [33], and promotes repair by enhancing mesenchymal stem cell homing [34], myelin regeneration, and modulating neuropathic pain via CXCR4 signaling [35]. The upregulation of CXCL12

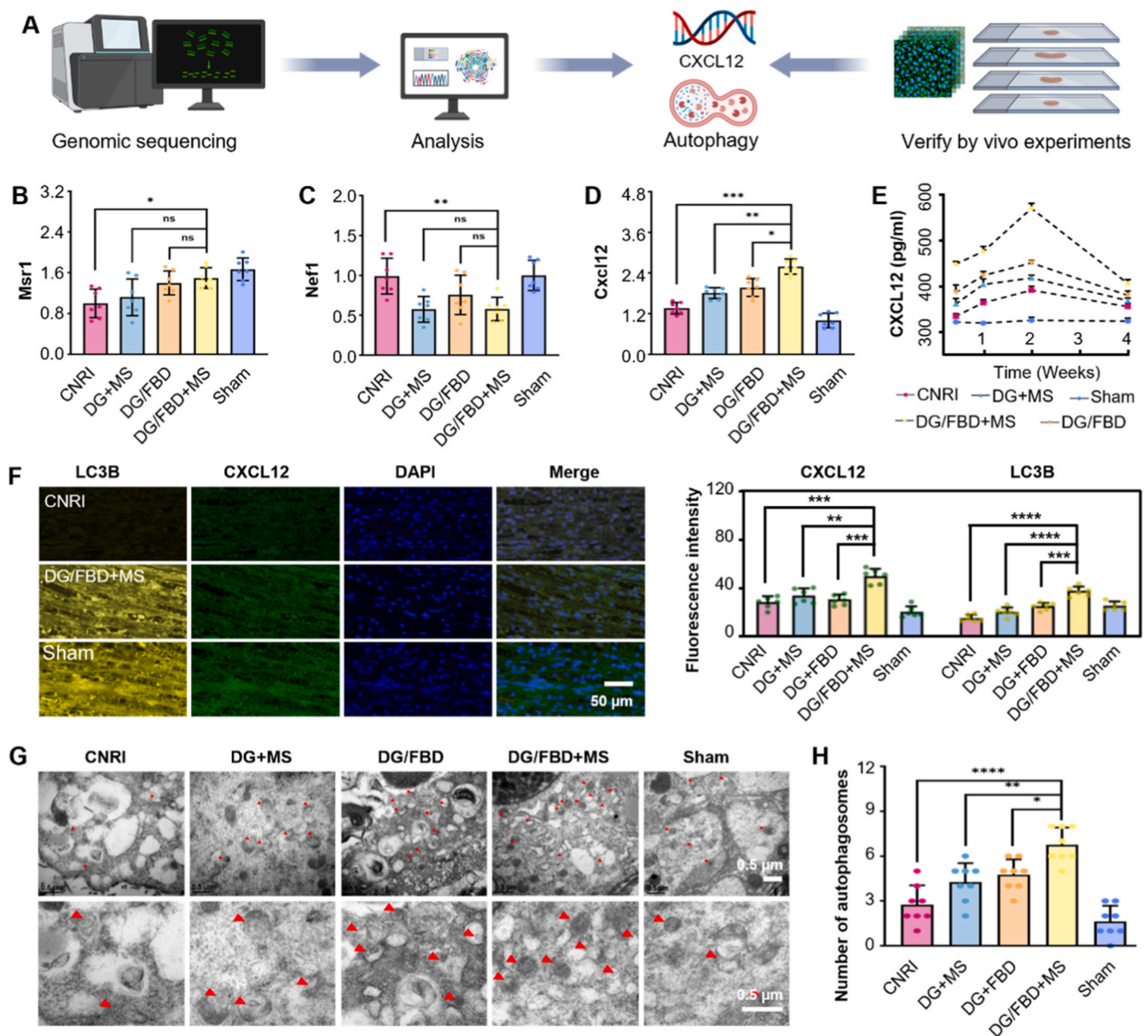


Fig. 6. Activation of the CXCL12-autophagy signaling axis. (A) Experimental validation strategy. (B–D) qRT-PCR analysis of Nefl, Msr1, Cxcl12. (E) CXCL12 protein expression levels. (F) Immunofluorescence co-localization and quantification of CXCL12 (green) and LC3B (yellow). (G) TEM images of autophagosomes. (H) Statistical quantification of autophagosome counts. The data are presented as mean $\pm$ SEM (* p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

in the DG/FBD + MS group supports its role as a critical mediator in CNRI repair. Immunofluorescence staining at 4 weeks post-surgery revealed significantly higher CXCL12 expression in DG/FBD + MS group (Fig. 6F), consistent with qPCR and ELISA results. Concurrently, autophagy marker LC3B (a well-established marker of autophagosome formation [36]) was upregulated (Fig. 6F and S25), while P62 (a critical indicator of autophagic flux) was markedly downregulated (Fig. S26). Previous studies have demonstrated that the upregulation of LC3B and

the downregulation of P62 are indicative of enhanced autophagy [37]. This differential expression pattern suggests an increase in autophagic flux in the DG/FBD + MS group. TEM further confirmed these observations by revealing an increased abundance of autophagosomes in the DG/FBD + MS group (Fig. 6G and H). Collectively, these findings indicate that the magnetic field-driven hydrogel spray simultaneously upregulates CXCL12 expression and promotes autophagic flux. While the potential correlation between these two phenomena warrants

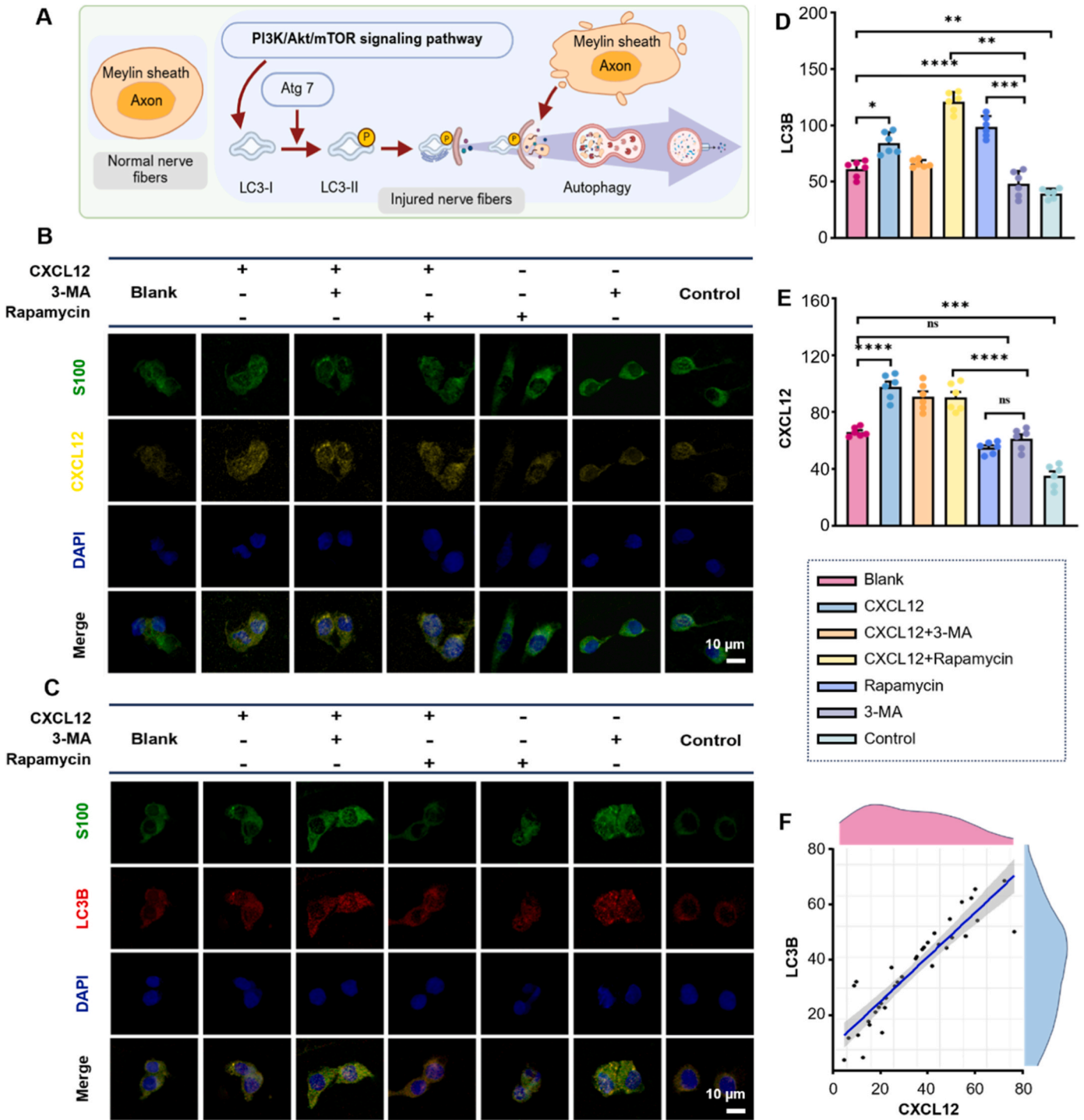


Fig. 7. CXCL12 enhances Schwann cell autophagy. (A) Proposed autophagy-mediated repair mechanism in Schwann cell. (B–C) Immunofluorescence staining of S100 (green), CXCL12 (yellow), LC3B (red), and DAPI (blue). (D–E) Immunofluorescence intensity analysis of LC3B and CXCL12 quantified from immunofluorescence images in Fig. 7B and C. (F) Correlation analysis between CXCL12 and LC3B. Data presented as mean $\pm$ SEM (ns: not significant, $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

further investigation, these results highlight the dual role of the hydrogel in enhancing both CXCL12 expression and autophagic activity. Notably, previous studies have shown that magnetoelectric stimulation promotes nerve repair through various mechanisms, including the modulation of the MAPK/ERK/PI3K pathway, the influence on intracellular ion channel activity, and the enhancement of cell migration [38–40]. However, our findings differ in that we have innovatively discovered that magnetoelectric stimulation can upregulate CXCL12

expression and promote autophagic flux. This novel discovery provides new insights into how magnetoelectric stimulation simultaneously regulates chemokine signaling and autophagic activity to enhance nerve repair. Considering that magnetoelectric hydrogels may still be present after 4 weeks, we acknowledge that extended evaluation periods are needed to explicitly evaluate the chronic effects of hydrogel-mediated autophagy on neuronal survival, glial homeostasis, and functional recovery, particularly in the context of material persistence and potential

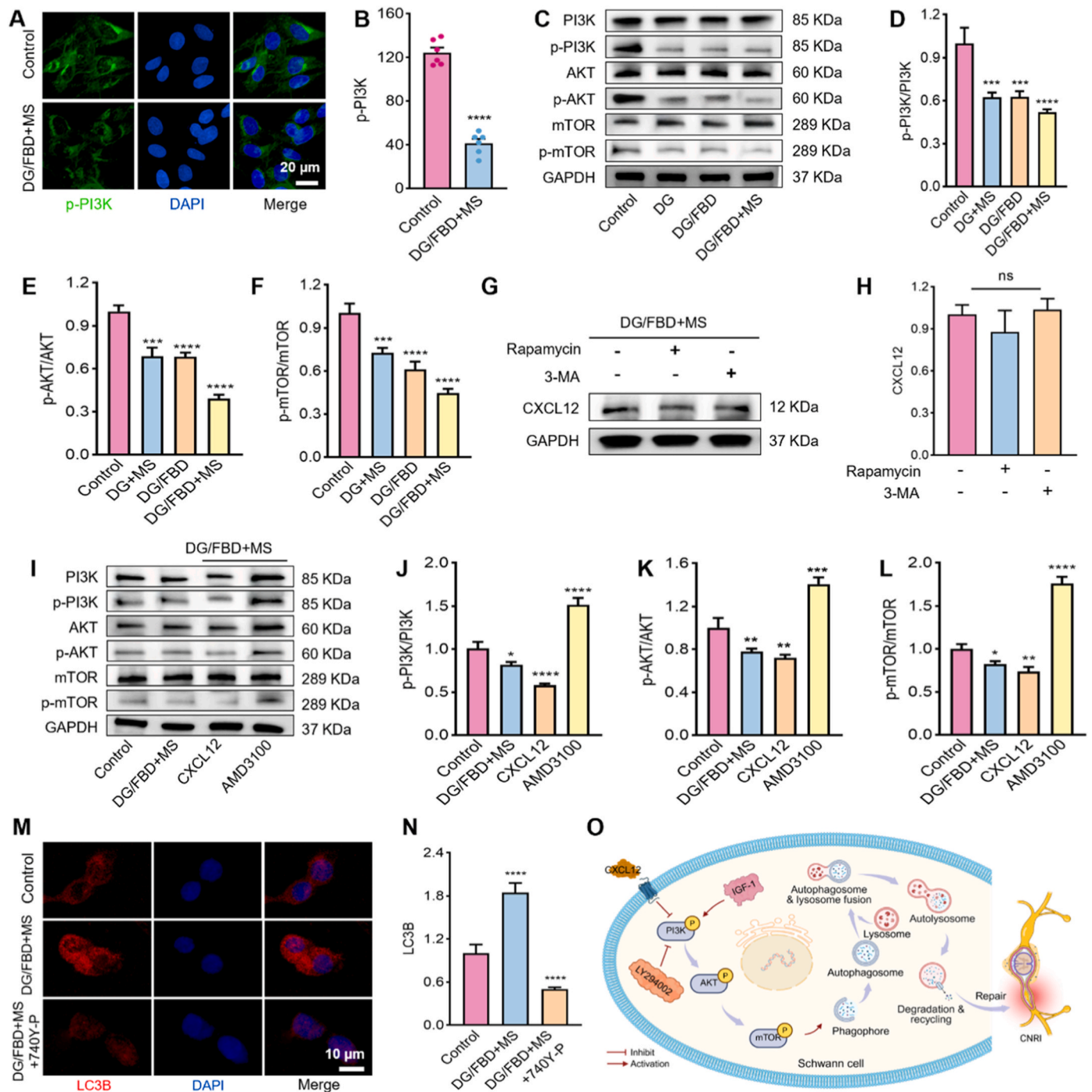


Fig. 8. CXCL12 enhances Schwann cell autophagy via PI3K/AKT/mTOR pathway. (A) Immunofluorescence image of PI3K. (B) Quantified fluorescence intensities of PI3K. (C) Protein band plots of PI3K, p-PI3K, AKT, p-AKT, mTOR, and p-mTOR (loading control: GAPDH). (D–F) Relative protein expressions of p-PI3K/PI3K, p-AKT/AKT, p-mTOR/mTOR. (G) Protein band plots of CXCL12; (H) Relative protein expressions of CXCL12. (I) Protein band plots of PI3K, p-PI3K, AKT, p-AKT, mTOR, and p-mTOR (loading control: GAPDH). (J–L) Relative protein expressions of p-PI3K/PI3K, p-AKT/AKT, p-mTOR/mTOR. (M) Immunofluorescence image of LC3B. (N) Quantified fluorescence intensities of LC3B. (O) Therapeutic mechanism of CXCL12-mediated autophagy. Data presented as mean $\pm$ SEM (* p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001, compared with Control group).

compensatory adaptations.

2.7. CXCL12 modulates autophagy enhancement in schwann cells via the PI3K/AKT/mTOR signaling pathway

We then questioned if there is a correlation between CXCL12 and autophagy. Schwann cells (SCs) play a pivotal role in peripheral nerve injury repair by migrating to the injury site, clearing myelin debris, and facilitating autophagosome formation, which creates a conducive environment for nerve regeneration (Fig. 7A) [41–46]. Based on this, we selected SCs as the cell model to investigate whether CXCL12 directly enhances autophagy under the action of magnetic field-driven DG/FBD hydrogel. The specific process is to treat SCs with exogenous CXCL12 overexpression, autophagy inhibitor 3-MA, and autophagy agonist Rapamycin in vitro experiments.

Immunofluorescence images (Fig. 7B and C) and quantitative analysis (Fig. 7D and E) revealed significant differences between the Blank, Rapamycin, and 3-MA groups, confirming that the agonist and inhibitor effectively modulated SCs activity and validating the experimental conditions. Compared with the blank group and the control group, the magnetic field driven DG/FBD hydrogel significantly increased the expression of CXCL12 and the occurrence of autophagy (Fig. 7D and E). Notably, the CXCL12-treated group exhibited enhanced autophagic flux compared to the Blank group, while the CXCL12+ Rapamycin group demonstrated greater autophagic activity than Rapamycin alone. Furthermore, the CXCL12+3-MA group partially reversed 3-MA-induced autophagy inhibition. Correlation analysis using Schwann cell marker S100 as reference revealed increasing CXCL12 expression corresponded to elevated autophagy marker LC3B (Fig. 7F), indicating CXCL12 promotes autophagic processes.

Previous studies have highlighted the positive regulation of autophagy through inhibition of the PI3K/AKT/mTOR signaling pathway [36,47]. However, detailed studies on the mechanisms of autophagy regulation in magnetoelectric nerve modulation are lacking. Our gene set enrichment analysis (GSEA) suggests that this pathway is involved in Schwann cells (SCs) treated with the magnetic field-driven DG/FBD hydrogel. Immunofluorescence staining revealed that SCs exposed to DG/FBD hydrogel exhibited reduced levels of p-PI3K (Fig. 8A and B). Based on these findings, we hypothesized that the hydrogel enhances SCs autophagy by modulating the PI3K/AKT/mTOR signaling pathway via CXCL12.

To test this hypothesis, Western blot analysis was performed to assess key signaling proteins, including PI3K, AKT, mTOR, p-PI3K, p-AKT, and p-mTOR (Fig. 8C). The results showed a significant decrease in the expression of p-PI3K, p-AKT, and p-mTOR in SCs treated with the DG/FBD hydrogel (Fig. 8D–F), indicating that the hydrogel attenuates the PI3K/AKT/mTOR pathway. Next, we evaluated whether autophagy influences CXCL12 expression. Western blot analysis revealed no significant changes in CXCL12 expression levels induced by autophagy (Fig. 8G and H). Furthermore, the DG/FBD hydrogel modulated the PI3K/AKT/mTOR pathway through CXCL12 regulation (Fig. 8I–L). The application of the PI3K inhibitor 740Y-P confirmed that the hydrogel promotes autophagy through inhibition of the PI3K/AKT/mTOR pathway (Fig. 8M and N).

These findings support our hypothesis that CXCL12 inactivates the PI3K/AKT/mTOR pathway, thereby enhancing Schwann cell autophagy (Fig. 8O). While prior sporadic studies suggest CXCL12 may promote SCs migration and autophagy via the PI3K-AKT-mTOR signaling pathway [48,49], our work provides innovative confirmation of this mechanism in magnetoelectric hydrogel-mediated nerve regeneration. These findings expand the understanding of Schwann cell-mediated repair mechanisms and open new therapeutic avenues for enhancing functional recovery post-injury.

Beyond CNRI repair, the DG/FBD hydrogel's rapid gelation, robust adhesion, and anatomical adaptability making it potential for minimally invasive neuromodulation in deep and complex nerve injuries (e.g.,

sciatic transection, brachial plexus, facial nerve). While magnetoelectric principle is universal, the downstream biology depends on the local setting. Prior studies implicated calcium signaling, MAPK/ERK activation, or neurotrophic secretion as key drivers of regeneration, yet the dominant pathway is dictated by material chemistry, stimulation parameters, and cellular context [50–52]. Here, we uniquely link CXCL12-autophagy a signaling to Schwann cell remodeling, emphasizing context-dependent biological outcomes and the need for tailored mechanistic investigation across different injury models.

Moreover, this magnetoelectric hydrogel platform offers distinct practical advantages in anatomically confined regions such as the cervical nerve root. Unlike conventional electrical stimulation, which requires invasive electrode placement and struggles to interface with branched nerves, the wireless sprayable design provides conformal, lead-free stimulation [53–55]. Compared with other wireless modalities, magnetoelectric coupling overcomes limitations of photoresponsive (poor tissue penetration), piezoelectric (insufficient endogenous strain), and ultrasound-driven (bone-attenuated signal) systems [13,56–58]. These advantages, amplified by the newly identified CXCL12-autophagy axis, collectively support the unique therapeutic potential of the DG/FBD + MS system. To reinforce clinical application potential, future studies will implement serial in vivo hydrogel tracking to comprehensively evaluate long-term hydrogel stability, bioresorption and biosafety.

3. Conclusion

This study establishes a wireless magnetoelectric hydrogel spray (DG/FBD) as an innovative therapeutic platform for CNRI repair, synergistic integrating biocompatible, anatomical adaptability, and wireless neurostimulation. The rapid sol-gel transition and sprayable fluidity, enable precise deposition of hydrogel solution along tortuous nerve roots, achieving strong bioadhesion while conforming to branched architectures of cervical root. Both in vitro and in vivo evaluations demonstrate that DG/FBD-mediated magnetoelectric stimulation significantly enhances neural differentiation, axonal growth, and myelination. Notably, it leads to improvements in sensory functions, including pain and temperature sensation, as well as a significant enhancement in grip strength. Mechanistically, we pioneer the discovery that magnetoelectrically triggered CXCL12 upregulation initiates Schwann cell remodeling through PI3K/AKT/mTOR pathway inhibition and autophagic flux potentiation, bridging bioelectronic signaling with metabolic clearance. These findings not only offer an adaptive wireless magnetoelectric stimulation strategy tailored to complex anatomical niches, but also establish CXCL12 as a potential therapeutic target for peripheral neuropathy.

4. Methodology

4.1. Chemical reagents

The following chemicals were utilized in this study: FeCl₃·6H₂O, Ba(OH)₂·8H₂O, tetrabutyl titanate (TBOT), 3-Aminobenzenboronic acid, and dopamine hydrochloride (DA-HCl), which were sourced from Aladdin (Shanghai, China). Dextran (MW: 70,000), 2-hydroxy-1-[4-(hydroxyethoxy) phenyl]-2-methyl-1-propanone (I2959), Methacrylic anhydride (MA, >94 %), and Type A porcine gelatin were obtained from Sigma-Aldrich (USA). Sodium periodate (NaIO₄, >99 %) was purchased from Adamas (China). Additional reagents, including ethylene glycol (EG), sodium citrate (Na₃CT), anhydrous ethanol, acetonitrile (ACN), ammonium acetate (NH₄OAc), sodium periodate (NaIO₄), and 30 % ammonia solution, were provided by Chengdu Kelong Reagent Co., Ltd. (China). All chemicals used were of analytical grade and applied without further purification.

4.2. Synthesis of $\text{Fe}_3\text{O}_4@\text{BaTiO}_3@\text{DA}$ (FBD) core-shell nanoparticles

Fe_3O_4 (FO) nanoparticles are synthesized via a solvothermal method. $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ (1.350 g), Na_3CT (0.4 g), and NH_4Ac (3.854 g) are dissolved in ethylene glycol (70 mL), stirred at 25 °C for 1 h, and reacted at 200 °C for 12 h under high pressure. The product is magnetically collected, washed, and dispersed in ethanol. Subsequently, the FO suspension (5 ml) is mixed with ethanol (85 ml)/acetonitrile (30 ml), ultrasonicated, and reacted with TBOT for 1.5 h, followed by magnetic collection and washing to obtain FO@TO (FT). The FT suspension is further ultrasonically dispersed, mixed with $\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$ solution (0.015 g/mL) and stirred for 15 min, followed by the addition of ammonia (2 ml) and hydrothermal reaction at 200 °C for 8 h. The product is washed with formic acid/water and magnetically collected to yield FO@BTO (FB) [59]. Finally, FB is mixed with Tris-HCl buffer (10 mM, pH 8.5) for 10 min, and dopamine (DA, 1.2 g) is added and continuously stirred at 25 °C for 24 h. The product is washed three times with distilled water and collected using a neodymium magnet to obtain FO@BTO@DA (FBD).

4.3. Characterizations of nanoparticles

The morphology of the nanoparticles was examined using field emission scanning electron microscopy (FESEM, S-4800, Hitachi, Japan) and transmission electron microscopy (TEM, JEM-2100F, JEOL). To determine the elemental composition, energy-dispersive spectroscopy (EDS) analysis was performed on the TEM system (JEM-2100F, JEOL). The zeta potential of the nanoparticles was measured through dynamic light scattering (DLS) using a Zetasizer Nano ZS90 (Malvern Instruments). Phase identification was conducted using an X-ray diffractometer (AXIS Ultra DLD, $\text{Cu K}\alpha = 1.5406 \text{ \AA}$). Magnetic characteristics were evaluated using a vibrating sample magnetometer (VSM, model BHV-525, Riken Denshi, Japan). Atomic force microscopy (AFM) and piezoresponse force microscopy (PFM) were carried out using a Bruker Dimension Icon system with conductive platinum-iridium silicon cantilevers (SCM-PIT, Bruker). The local polarization switching curves for the ferroelectric BaTiO_3 ceramics (FBD) were recorded within a voltage range of -12 V to $+12 \text{ V}$, under the application of magnetic field stimulation at 1000 Oe.

Finite Element Simulation. The finite element method (FEM) simulation of the magnetoelectric properties of FBD nanoparticles is conducted using COMSOL Multiphysics, following methodologies established in previous literature. The magnetoelectric phenomena are characterized by the interplay between magnetic fields, solid mechanics, and electrostatics. A pair of energized coils generated a magnetic field with a strength of 15 mT, oriented along the Y-axis. The surrounding environment is modeled as air, with relative permittivity and relative magnetic permeability both set to 1. For this simulation, the material parameters used are the default values provided within COMSOL Multiphysics.

4.4. Preparation of DG/FBD hydrogel

Phenylboronic acid-grafted oxidized dextran (PBA-OD) and gelatin acrylamide (MG) are synthesized following established protocols from previous literature [60,61]. First, the oxidation-reduction reaction with sodium periodate is used to obtain OD. Subsequently, PBA-OD is synthesized through a Schiff base reaction. In addition, MG is synthesized by grafting methacrylic anhydride (MA) onto the gelatin molecular chain via esterification modification. The successful synthesis of PBA-OD and MG is confirmed using ^1H NMR spectroscopy (AV II-600 MHz, Bruker) and FTIR spectroscopy (Thermo Fisher Nicolet Is10). The DG/FBD hydrogel is prepared using the following formulated solutions: FBD (0.5 % w/v), PBA-OD (20 % w/v), and MG (30 % w/v). Transfer solution A (containing PBA-OD and FBD) and solution B (containing MG) into two separate spray bottles. Apply the solutions in

an alternating sequence: spray solution A onto the target surface, followed immediately by solution B. Repeat this alternating spray sequence to obtain the sprayable DG/FBD hydrogel. Place the prepared DG/FBD hydrogel under UV irradiation (wavelength: 365 nm, intensity: 3 W/cm^2) for 20 s using the OmniCure® S1500 EXPO system to complete the crosslinking of the DG/FBD hydrogel.

4.5. Characterizations of hydrogel

The incorporation of FBD into the hydrogel matrix was confirmed using UV-visible spectroscopy at a wavelength of 295 nm. The morphology of the lyophilized hydrogels was analyzed using scanning electron microscopy (SEM). Rheological measurements of the hydrogels were carried out using a modular compact rheometer (MCR-302, Anton Paar). To evaluate hydrogel performance in physiologically relevant conditions, gelation kinetics and shear thinning behavior were measured at 37 °C in PBS (simulating tissue moisture), whole blood, and RO water using rheometry. Adhesion properties of the hydrogels were assessed using a universal testing machine. Furthermore, to evaluate hydrogel adhesion under physiologically relevant neural-tissue conditions, an ex-vivo nerve-adhesion assay was performed. Briefly, the exposed cervical nerve root (C7) in a rat model was spray-coated in situ with DG/FBD hydrogel. The coated nerve was then explanted and subjected to gentle PBS flushing (1 min) under visual inspection. Hydrogel retention was further verified via H&E staining of longitudinal nerve sections, which allowed qualitative assessment of hydrogel-tissue interfacial integrity. Additionally, an ex-vivo adhesion assay was conducted by spraying hydrogel layers on freshly isolated rat cervical nerve roots (C7). Hydrogel thickness and homogeneity were verified via H&E staining of longitudinal nerve sections using NDP.view2 software to evaluate thickness and homogeneity. The distribution of micro-Young's modulus within the hydrogels was characterized with a Piuma Chiaro Nanoindenter (Optics11). Magnetic properties were evaluated using a vibrating sample magnetometer (VSM, model BHV-525, Riken Denshi, Japan). The details were shown in Supporting Information. (Methodology S1)

4.6. Cell culture

Bone Marrow Stromal Cells (BMSCs) were isolated from the femora and tibiae of 4-week-old Sprague-Dawley (SD) rats. Following aseptic removal of the bones, bone marrow was flushed out using phosphate-buffered saline (PBS) with a sterile syringe. The collected bone marrow cells were subjected to centrifugation at 1000g for 5 min. The resulting cell pellet was resuspended in α -MEM (Gibco, Gaithersburg, MD, USA), supplemented with 10 % fetal bovine serum (FBS; Gibco, Carlsbad, CA, USA) and 1 % penicillin-streptomycin (Thermo Fisher Scientific). The cells were incubated at 37 °C in a humidified atmosphere containing 5 % CO_2 for 48 h. Non-adherent cells were discarded, while adherent cells were cultured until they reached approximately 80 % confluence, defined as passage one (P1). For experimental purposes, passage three (P3) BMSCs were utilized.

PC12 cells, obtained from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China), were maintained in high-glucose Dulbecco's Modified Eagle Medium (DMEM; Hyclone, China) supplemented with 10 % FBS (TBD Science, China) and 1 % penicillin-streptomycin (Hyclone, China). Cells were kept in a 37 °C incubator with 5 % CO_2 and 95 % humidity. To induce differentiation, PC12 cells were treated with nerve growth factor (NGF).

Primary culture of Schwann cells was prepared according to previously published methods [62,63]. Briefly, Schwann cells were isolated from the sciatic nerves of 48-h-old Sprague-Dawley rats. To suppress the rapid proliferation of fibroblasts, the Schwann cells were cultured for one day in a purification medium, after which they were transferred to a medium without growth factors for a one-day recovery period. Finally, the cells were cultured in a medium containing Schwann cell growth

factors for further cultivation. Cells were maintained at 37 °C in a humidified incubator with 5 % CO₂ and 95 % humidity. For optimal proliferation, the culture medium was refreshed every 2–3 days, and cells were passaged upon reaching 80–90 % confluence using 0.25 % trypsin-EDTA for detachment. According to the experimental design, Schwann cells were cultured after treatment with or without CXCL12 (100 ng/mL, Beyotime, China), 3-MA (100 ng/mL, Beyotime, China) and Rapamycin (20 µmol/L, Beyotime, China).

4.7. Cell viability and proliferation assay

BMSCs and PC12 cells were seeded into 24-well plates at a density of 1×10^4 cells per well and co-cultured with respective hydrogel constructs. Magnetic field stimulation was applied using a magnetic field generator equipped with circular Helmholtz coils. Hydrogels with cell seeding were positioned at the center of the coils, where they were exposed to a pulsed magnetic field (15 mT, 60 Hz, duty cycle of 50 %) generated by applying a pulsed current. This stimulation protocol was administered daily for 30 min, and all the parameter set was selected from prior optimization as the optimal balance of efficient magneto-electric conversion, biosafety, and negligible heating [64–67]. Cell viability was assessed on days 2, 5, and 8. Cells were stained with fluorescein diacetate (FDA, 1 µg/mL) and propidium iodide (PI, 1 µg/mL), and images were captured using a confocal laser scanning microscope (CLSM, Zeiss LSM880, Germany).

4.8. Cell morphology observation and measurement

For the observation of cell morphology, the samples were first incubated with a 50 µg/mL solution of Alexa Fluor-488 phalloidin (Invitrogen) at room temperature. Afterward, DNA was counterstained with DAPI (Sigma) to enable imaging. The lengths of PC12 cells ($n \geq 60$) were measured using ImageJ and Origin software, where cell length was defined as the distance between the tip of a neurite and its opposite end.

4.9. Construction of animal models

All animal procedures adhered strictly to the ARRIVE guidelines (Animal Research: Reporting of In Vivo Experiments) and were approved by the Medical Ethics Committee of Sichuan University (Approval No.: KS2022863). Adult male Sprague-Dawley (SD) rats, weighing 300–350 g, were sourced from Chengdu Dazhuo Experimental Animal Company (Chengdu, China) and randomly allocated into five groups: CNRI, DG + MS, DG/FBD, DG/FBD + MS, and Sham. Prior to surgical procedures, rats were anesthetized with a 3.0 % pentobarbital sodium solution (1.5 mL/kg body weight), and anesthesia was maintained using isoflurane inhalation (4 % for induction, 2 % for maintenance). The left C7 cervical dorsal nerve root underwent transient compression injury as previously described [68–70].

For the surgery, rats were positioned in a prone posture, and a midline incision was made from the cranial base to the second thoracic vertebra. After careful dissection of muscle and soft tissues, the distal nerve plexus was exposed. A hemilaminectomy and partial facetectomy at the C6/C7 level were performed to access the left spinal cord and C7 dorsal nerve root. To fully expose the nerve root, a small incision was made at the dural sac where the dorsal root enters the spinal cord [69]. The Sham group only involved exposing the cervical nerve root without any further treatment. In the experimental groups, a microvascular clamp applied a 10 g compressive force to the left C7 dorsal nerve root for 15 min [70–72]. Post-compression, the CNRI group did not receive hydrogel treatment, whereas the remaining groups were treated with their respective hydrogels as detailed in Fig. S27A–H (Supporting Information). Surgical sites were closed using sutures and staples, and the animals were allowed to recover in a monitored environment under ambient air conditions. 24 h post-surgery, rats assigned to magnetic stimulation groups received daily treatment. A pair of custom Helmholtz

coils was positioned to surround the cervical region, through which a pulsed current was applied to generate a uniform pulsed magnetic field (15 mT, 60 Hz, 50 % duty cycle). Each stimulation session lasted 30 min and was administered once daily until the endpoint of the experiment. All the parameter set was selected from prior optimization [64–67].

4.10. Enzyme-linked immunosorbent assay (ELISA)

At post-surgical time points of 0, 1, 3, 7, 14, and 28 days, three rats from each of the Sham and injured groups were euthanized. A 0.8 cm segment of cervical nerve root tissue was excised from the injury site and immediately frozen for further analysis. For cytokine measurement, tissue lysates were prepared by homogenizing 50 mg of nerve tissue in RIPA buffer. The concentration of CXCL12 in the lysates was quantified using an ELISA kit (Meimian, Jiangsu, China), according to the manufacturer's instructions. Each sample was assayed at least twice, and the experiments were conducted independently with three biological replicates.

4.11. Immunofluorescence staining

Cells or tissue samples were fixed in 4 % paraformaldehyde for 20 min and subsequently permeabilized with 0.2 % Triton X-100 (Sigma-Aldrich, USA) for 15 min. Following permeabilization, the samples were blocked with 5 % bovine serum albumin (BSA, Biyuntian, China) for 1 h at room temperature. Primary antibodies against Tuj1 (Abcam, China), NF (Abcam, China), GFAP (Abcam, China), PSD95 (Abcam, China), MBP (Abcam, China), NF200 (Beyotime, China), S100 (Beyotime, China), CXCL12 (Beyotime, China), LC3B (Beyotime, China), and P62 (Beyotime, China) were applied and incubated overnight at 4 °C. The samples were then incubated with corresponding fluorescently labeled secondary antibodies at 37 °C for 1 h. For cytoskeletal staining, BMSCs were treated with fluorescein-labeled phalloidin at a concentration of 50 µg/mL. Nuclear counterstaining was performed using DAPI (10 µg/mL) for 5 min, followed by brief washes. Images were acquired using confocal laser scanning microscopy (CLSM).

4.12. Transmission electron microscopy (TEM) of nerve remyelination

Four weeks post-surgery, the injured cervical nerve roots were excised and immediately fixed in a cold solution of 0.5 % glutaraldehyde. Following fixation, the tissues were carefully trimmed, washed, and dehydrated through a graded series of ethanol solutions. Samples were then embedded in Epon 812 epoxy resin, and ultrathin sections (60 nm thick) were prepared for TEM analysis. The sections were stained with lead citrate and uranyl acetate to enhance contrast. Parameters indicative of nerve regeneration, including myelin thickness, myelin fiber diameter, and the number of myelin layers, were quantified using ImageJ software. Additionally, the G-ratio value (the ratio of axon diameter to the total fiber diameter) was determined from TEM of nerve cross-sections by manually outlining the axon and fiber contours and calculating the ratio.

4.13. Tissue processing and transcriptome sequencing

Male adult SD rats (7–8 weeks old, weighing 200–250 g) were randomly divided into five groups: CNRI, DG + MS, DG/FBD, DG/FBD + MS and Sham. All animals were maintained under conditions with standardized feed and free access to water, as detailed in the animal modeling section. On day 28 post-intervention, animals were systemically anesthetized, and cervical nerve root tissues were rapidly excised, immediately frozen in liquid nitrogen, and stored at –80 °C until further analysis. Total RNA was extracted using the Qiagen RNeasy kit, and its integrity was evaluated using a Nanodrop spectrophotometer and an Agilent 2100 Bioanalyzer. High-quality RNA samples were then sent to Novogene Company (Chengdu, China) for whole-genome transcriptome

sequencing. The sequencing data underwent quality control checks using FastQC, and low-quality reads along with adapter sequences were removed using Trimmomatic.

4.14. Bioinformatics analysis

Differential gene expression analysis was conducted using the R package "limma" (version 4.4.0) [73]. Differentially expressed genes (DEGs) between the Sham and DG/FBD + MS groups were identified, with results visualized through volcano plots. A threshold of $p < 0.05$ and $|\log_2 \text{fold change}| > 1$ was set to determine statistical significance. The "RobustRankAggreg" method was employed to refine DEG selection, ensuring minimized biases across multiple datasets. Gene Ontology (GO) analyses were performed using the ClusterProfiler package, with a significance level of $p < 0.05$ [74]. Gene Set Enrichment Analysis (GSEA) was also conducted to evaluate the enrichment of genes in significant functional pathways [75]. To identify key biomarkers from co-expressed genes for further analysis, we utilized the Boruta feature selection algorithm [76], combined with the Least Absolute Shrinkage and Selection Operator (LASSO) [77].

4.15. Histological and immunofluorescence staining of CNR

CNR samples were initially fixed in 4 % paraformaldehyde overnight at 4 °C, followed by cryoprotection in 30 % sucrose for 48 h. The tissues were then embedded in paraffin and sectioned into 5 µm thick slices. Hematoxylin and eosin (H&E) staining was performed to evaluate nerve injury and recovery. For immunofluorescence staining, the sections were incubated with primary antibodies against MBP, NF200, S100, CXCL12, LC3B, and P62. Fluorescent secondary antibodies, Alexa Fluor 488 and Alexa Fluor 555, were used for detection, and the nuclei were counterstained with DAPI.

4.16. Fluorescence intensity quantification

Fluorescence intensity was quantified using a laser scanning confocal microscope (CLSM, Zeiss LSM880, Germany). Acquired images were analyzed with ImageJ software (version 1.53t, NIH, USA). For both tissue sections and cultured cells, regions of interest (ROIs) were delineated manually or automatically based on DAPI + nuclei or specific immunostaining signals. Mean fluorescence intensity (MFI) was measured within each ROI and normalized either to the corresponding area (for tissue sections) or to the cell number (for cultured cells). Background intensity, measured from adjacent unstained or non-fluorescent regions, was subtracted from all values. For co-localization analysis, Pearson's correlation coefficient was calculated using the Coloc 2 plugin in ImageJ. All quantifications were performed on at least three independent biological replicates, with multiple fields of view analyzed per sample.

4.17. Von Frey and hargreaves sensitivity testing

Mechanical sensitivity was evaluated using Von Frey filaments within an acrylic chamber equipped with a stainless-steel mesh floor. Following a 20-min acclimatization period, mechanical withdrawal thresholds were determined using the up-down method. For thermal sensitivity testing, the Hargreaves apparatus was employed to deliver radiant heat stimuli, and the latency to paw withdrawal was recorded. A cutoff time of 30 s was used to prevent potential tissue damage.

4.18. Grip strength test

Grip strength of the forelimb on the operated side was assessed using a grip strength meter (YLS-13A, Jinan Yiyan Technology Development Co., Ltd., China). Each rat underwent three consecutive trials, and the mean value was calculated for statistical analysis.

4.19. Western blot analysis

Cellular proteins were extracted using RIPA lysis buffer supplemented with protease and phosphatase inhibitors. Protein concentrations were quantified using a BCA assay kit (Solarbio, China). Equal protein amounts (10 µg) were resolved by 10 % SDS-PAGE and transferred onto polyvinylidene fluoride (PVDF) membranes. The membranes were blocked with 5 % non-fat milk for 1 h at room temperature, followed by incubation overnight at 4 °C with primary antibodies targeting PI3K, p-PI3K, AKT, p-AKT, mTOR, p-mTOR, CXCL12, and GAPDH. After incubation with appropriate secondary antibodies, protein bands were detected using an enhanced chemiluminescence kit (Biyuntian, China). Band intensities were analyzed using ImageJ software.

4.20. Statistical analysis

Data are expressed as mean ± standard deviation (SD). Statistical significance was assessed using *t*-test and one-way ANOVA, followed by Tukey's post-hoc test for multiple comparisons. All statistical analyses were conducted using GraphPad Prism 9.5.0 (GraphPad Software Inc.). Significance levels were determined by (*) $p < 0.05$, (**) $p < 0.01$, (***) $p < 0.001$, and (****) $p < 0.0001$.

CRediT authorship contribution statement

Rongguo Yu: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jiamei Xiao:** Writing – original draft, Validation, Software, Formal analysis, Data curation. **Zhencheng Xiong:** Supervision, Software, Project administration, Formal analysis. **Xiaoyang Wu:** Validation, Supervision, Software, Methodology, Investigation, Formal analysis. **Chaoyi Zhang:** Software, Project administration, Methodology. **Yusheng Zhang:** Software, Project administration, Methodology. **Kangkang Huang:** Project administration, Methodology. **Jing Sun:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Conceptualization. **Hongsong Fan:** Writing – review & editing, Validation, Supervision, Funding acquisition, Data curation, Conceptualization. **Hao Liu:** Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biomaterials.2026.123983>.

Data availability

Data will be made available on request.

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